DETAILED TECHNICAL STUDY NOTES

ReDiffDet

Rotation-Equivariant Diffusion Model for Oriented Object Detection

Based on the CVPR 2025 paper by Jiaqi Zhao et al.

Mathematical foundations, derivations,
architecture, algorithms, experiments, and limitations

Part 2

Reverse diffusion, equivariance proofs,
architecture, and aligned decoder

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1 The Reverse Process: Learning to Recover Objects

The forward process is fixed. The reverse process is learned.

The goal is to recover a clean object representation from a noisy one while using the image to determine which objects are actually present.

The reverse transition is written as

$$\boxed{p_\theta(z_{t-1} \mid z_t) = \mathcal{N}\left(z_{t-1}; \mu_\theta(z_t, t), \Sigma_\theta(z_t, t)\right)}$$

This is Equation (11) of the paper.

Here:

- z_t is the current noisy state.
- t is the diffusion timestep.
- μ_θ is the predicted reverse-process mean.
- Σ_θ is the reverse-process covariance.
- θ represents the learned model parameters.

The neural network estimates how the noisy state should be refined. In ReDiffDet, the image features and the aligned decoder are essential to this prediction.

1.1 Why Is the Image Necessary?

Imagine that the noisy state contains a candidate Gaussian somewhere in an aerial image.

The Gaussian alone cannot determine whether the corresponding region contains a ship, a bridge, a building, or background.

The model must inspect the image features to infer the object.

Thus, the actual conceptual process is

$$\boxed{\text{Noisy geometric candidates} + \text{Image evidence} \longrightarrow \text{Refined object predictions}}$$

The diffusion state carries the candidate geometry, while the image features provide the semantic and visual evidence needed to refine it.

1.2 What Does the Neural Network Predict?

The paper describes a network

$$\epsilon_\theta(z_t, t)$$

and a reverse transition parameterized by

$$\mu_\theta(z_t, t).$$

The key mathematical requirement for rotation equivariance is that the predicted mean transforms consistently under rotation.

The paper's proof explicitly states the condition

$$\boxed{Q\mu_\theta(z_t, t) = \mu_\theta(Qz_t, t)}$$

This means that rotating the current noisy state and then predicting the reverse mean should give the same result as predicting the mean first and rotating that prediction afterward.

This condition is central to the theoretical contribution of ReDiffDet.

2 The Central Theoretical Contribution: Proving Rotation Equivariance

The paper's theory is organized around three propositions:

1. The terminal standard Gaussian distribution is rotation-invariant.
2. The reverse transition is rotation-equivariant if its predicted mean is equivariant.
3. The final distribution is rotation-invariant when the terminal distribution and every reverse transition satisfy the required properties.

Let's derive these results carefully.

2.1 Proposition 1: Standard Gaussian Noise Is Rotation-Invariant

At the terminal timestep, ReDiffDet starts from

$$z_T \sim \mathcal{N}(0, I).$$

Consider a rotation $Q \in SO(2)$, and define

$$z'_T = Qz_T.$$

We need to show that z'_T has the same distribution as z_T.

First, calculate its mean:

$$\begin{aligned}\mathbb{E}[z'_T] &= \mathbb{E}[Qz_T] \\ &= Q\mathbb{E}[z_T] \\ &= Q0 \\ &= 0.\end{aligned}$$

Next, calculate its covariance:

$$\begin{aligned}\mathrm{Cov}(z'_T) &= \mathrm{Cov}(Qz_T) \\ &= Q\,\mathrm{Cov}(z_T)Q^\top \\ &= QIQ^\top \\ &= QQ^\top \\ &= I.\end{aligned}$$

Therefore,

$$z'_T \sim \mathcal{N}(0, I).$$

Hence,

$$\boxed{Qz_T \overset{d}{=} z_T}$$

The symbol $\overset{d}{=}$ means equality in distribution.

Why does this work?

Because the covariance is isotropic. There is no preferred direction in the standard Gaussian distribution, so rotation cannot change its shape.

This is Proposition 1 and Equations (9)–(10) on page 4 of the paper.

2.2 Proposition 2: The Reverse Transition Is Equivariant

Now consider the reverse-process distribution

$$p_\theta(z_{t-1} \mid z_t) = \mathcal{N}\left(z_{t-1}; \mu_\theta(z_t, t), \sigma_t^2 I\right),$$

using the isotropic reverse covariance assumed in the paper's proof.

We want to establish

$$\boxed{p_\theta(Qz_{t-1} \mid Qz_t) = p_\theta(z_{t-1} \mid z_t)}$$

The important assumption is

$$\boxed{\mu_\theta(Qz_t, t) = Q\mu_\theta(z_t, t)}$$

Let's derive the result.

The Gaussian density on the left is

$$p_\theta(Qz_{t-1} \mid Qz_t) = \frac{1}{2\pi\sigma_t^2} \times \exp\left[-\frac{\|Qz_{t-1} - \mu_\theta(Qz_t, t)\|^2}{2\sigma_t^2}\right].$$

By the assumed equivariance of the mean,

$$\mu_\theta(Qz_t, t) = Q\mu_\theta(z_t, t).$$

Substitute this into the density:

$$p_\theta(Qz_{t-1} \mid Qz_t) = \frac{1}{2\pi\sigma_t^2} \times \exp\left[-\frac{\|Qz_{t-1} - Q\mu_\theta(z_t, t)\|^2}{2\sigma_t^2}\right].$$

Factor out the rotation:

$$\begin{aligned}&\|Qz_{t-1} - Q\mu_\theta(z_t, t)\|^2 \\ &= \|Q\left(z_{t-1} - \mu_\theta(z_t, t)\right)\|^2.\end{aligned}$$

Since

$$Q^\top Q = I,$$

we have

$$\begin{aligned}\|Qx\|^2 &= (Qx)^\top (Qx) \\ &= x^\top Q^\top Q x \\ &= x^\top x \\ &= \|x\|^2.\end{aligned}$$

Therefore,

$$\begin{aligned}p_\theta(Qz_{t-1} \mid Qz_t) &= \frac{1}{2\pi\sigma_t^2} \times \exp\left[-\frac{\|z_{t-1} - \mu_\theta(z_t, t)\|^2}{2\sigma_t^2}\right] \\ &= p_\theta(z_{t-1} \mid z_t).\end{aligned}$$

Thus,

$$\boxed{p_\theta(Qz_{t-1} \mid Qz_t) = p_\theta(z_{t-1} \mid z_t)}$$

The proof rests on two facts:

- The predicted mean transforms equivariantly.
- The isotropic Gaussian noise distribution is unchanged by rotation.

The first fact is the difficult part in practice. The paper uses rotation-aware features and aligned region extraction to help the model satisfy it.

A mathematical qualification: if the reverse covariance were a general matrix rather than $\sigma_t^2 I$, the proof would require the covariance to transform compatibly under rotation as well.

2.3 Proposition 3: The Complete Reverse Process Preserves Rotational Symmetry

The next question is whether equivariance at individual steps carries through the entire diffusion chain.

The terminal distribution is rotation-invariant:

$$p(z_T) = p(Qz_T).$$

The reverse transitions satisfy

$$p_\theta(Qz_{t-1} \mid Qz_t) = p_\theta(z_{t-1} \mid z_t).$$

The joint reverse-process distribution factors as

$$\boxed{p_\theta(z_{0:T}) = p(z_T) \prod_{t=1}^{T} p_\theta(z_{t-1} \mid z_t)}$$

To obtain the final distribution, integrate out all intermediate states:

$$p_\theta(z_0) = \int p(z_T) \prod_{t=1}^{T} p_\theta(z_{t-1} \mid z_t)\, dz_{1:T}.$$

Now rotate the final state. Applying the same transformation consistently to the trajectory and using the two established symmetry properties gives

$$\begin{aligned} p_\theta(Qz_0) &= \int p(Qz_T) \prod_{t=1}^{T} p_\theta(Qz_{t-1} \mid Qz_t)\, dz_{1:T} \\ &= \int p(z_T) \prod_{t=1}^{T} p_\theta(z_{t-1} \mid z_t)\, dz_{1:T} \\ &= p_\theta(z_0). \end{aligned}$$

The change of variables introduces no extra volume factor because an orthogonal rotation has absolute determinant one.

Therefore,

$$\boxed{p_\theta(Qz_0) = p_\theta(z_0)}$$

This is the paper's Proposition 3, Equation (13).

There is a subtle but essential distinction here. The equality of probability densities under a common rotation is a statement about the symmetry of the distribution. It does not mean that a particular deterministic box prediction must remain unchanged after rotating the image. The box prediction should rotate with the image; the probability law of the properly transformed process has the symmetry established by the proof.

Also, the proof is conditional on the model satisfying the required equivariance assumptions. It does not establish that an arbitrary neural network automatically becomes equivariant merely because its output is modeled with a Gaussian. The architecture is designed to support that property.

3 How ReDiffDet Makes the Theory Practical

The proofs tell us what properties the model should satisfy. The architecture must now implement them while solving real object detection.

Figure 3 on page 4 presents the overall framework. It contains the image backbone and feature pyramid, the conditional encoder, Gaussian modeling, the diffusion and reverse processes, and the aligned decoder.

The two additional components beyond the diffusion formulation are especially important: conditional box generation and aligned feature extraction.

3.1 The Image Backbone and Feature Pyramid Network

The input image first passes through a backbone network and a feature pyramid network (FPN).

Let

$$I \in \mathbb{R}^{H\times W\times 3}$$

represent the input image.

The backbone extracts visual features at multiple resolutions:

$$F_{\text{backbone}} = \text{Backbone}(I).$$

The FPN combines information across resolutions:

$$F = \text{FPN}(F_{\text{backbone}}).$$

Why multiple resolutions?

- High-resolution features preserve information about small objects.
- Lower-resolution features capture broader context.
- Combining them helps detect objects of different sizes.

The paper's default rotation-aware configuration uses ReResNet-50 with ReFPN. It also evaluates ordinary ResNet-50 and Swin-T backbones.

The backbone is not itself the entire detector. It produces the visual representation used by the conditional encoder and the aligned decoder.

3.2 The Conditional Encoder

The paper observes that starting with completely unconstrained random boxes is problematic.

Imagine an image containing 100 ships, some only a few pixels wide. If the model initializes hundreds of candidate boxes randomly over the entire image, many may initially land on empty water. Others may have completely inappropriate sizes.

A diffusion model can refine noisy candidates, but it still needs useful visual information to identify which candidates should become objects.

ReDiffDet therefore introduces a conditional encoder.

The paper describes two components:

- A box subnet that proposes candidate boxes.
- A score subnet that scores those candidates.

Both use convolutional layers. The highest-scoring boxes are selected as conditional boxes.

Conceptually,

$$\boxed{B_{\text{cond}} = \text{TopK}\left(\text{BoxSubnet}(F), \text{ScoreSubnet}(F)\right)}$$

This expression summarizes the procedure rather than reproducing a specific implementation-level equation.

The conditional encoder provides an image-dependent prior for the locations of potential objects.

Why Is Conditioning Useful? It reduces the search space. Instead of treating every location and scale in the image as equally plausible, the model begins with candidates that already have some visual support.

The conditional boxes also provide centers for the Gaussian distributions used during inference.

However, conditioning does not guarantee that every candidate is correct. The decoder must still refine boxes and classify them, and an imperfect conditional encoder can omit true objects or produce false candidates.

3.3 How Conditional Boxes Influence the Gaussian

The paper uses the conditional boxes differently during training and inference.

During training:

1. Ground-truth boxes are converted to Gaussian distributions.
2. Conditional boxes are also converted to Gaussian distributions.
3. Ground-truth Gaussian distributions are padded to a fixed number of instances.
4. Samples are drawn and corrupted with diffusion noise.

During inference, the conditional box centers provide the origins for the initial Gaussian states, while the box dimensions control the Gaussian scale.

The paper describes this in Section 3.3 and Algorithms 1 and 2.

The reason for using conditional boxes is not to eliminate diffusion. It is to give diffusion more useful starting information.

4 The Aligned Decoder: Connecting Geometry to Image Features

This is the second major architectural contribution.

A candidate box is a geometric object. To classify it correctly, the model must extract the image features that belong to that object.

But a conventional feature map is arranged on a horizontal grid, while the candidate box may be tilted.

If the detector extracts features from the wrong region or uses poorly aligned features, the representation can mix object pixels with background pixels.

ReDiffDet addresses this with an aligned decoder consisting of three main operations:

1. RiRoI Align.
2. Self-attention.
3. Instance interaction.

The decoder is stacked in multiple layers. The paper's ablation study finds that four layers work best among the tested configurations.

4.1 RiRoI Align

RiRoI Align is a rotated-region feature extraction and alignment mechanism inherited from ReDet.

Suppose a candidate box has parameters

$$b = (c_x, c_y, w, h, a).$$

A point (u, v) in the box's local coordinate system can be mapped into the image coordinate system using

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c_x \\ c_y \end{bmatrix} + R_a \begin{bmatrix} u \\ v \end{bmatrix},$$

where

$$R_a = \begin{bmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{bmatrix}.$$

This mapping tells us where each point in the rotated region lies in the original feature map.

The alignment mechanism samples the appropriate feature locations and constructs a region representation whose coordinates follow the box.

For example, suppose the box surrounds a diagonal ship. A conventional horizontal crop may contain large amounts of water. A rotated crop follows the ship's long axis, making it easier to isolate the relevant visual features.

The paper's use of RiRoI Align serves two purposes:

- Align features with the geometry of the candidate.
- Produce rotation-invariant region features for classification and subsequent interaction.

The exact sampling implementation belongs to the inherited RiRoI Align method; the coordinate mapping above explains the geometry underlying it.

4.2 Self-Attention

After extracting region features, the decoder uses self-attention to model interactions among candidate instances.

Suppose there are N candidate boxes, each represented by a feature vector:

$$X \in \mathbb{R}^{N \times d}.$$

The standard self-attention operation is

$$Q = XW_Q, \qquad K = XW_K, \qquad V = XW_V,$$

followed by

$$\boxed{\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V}$$

Here d_k is the key dimension.

The matrix $QK^\top$ measures compatibility between candidate instances. Softmax converts these scores into attention weights.

Why can this help in aerial detection?

- Several nearby ships may have similar visual appearances.
- Neighboring aircraft may overlap in a feature map.
- Dense vehicle arrangements require the detector to distinguish individual instances.
- Candidates may provide useful contextual information about nearby objects.

Self-attention allows each candidate's representation to incorporate information from other candidates.

This equation describes the standard self-attention operation that explains the mechanism; it is not a claim that the paper introduces a new attention formula.

4.3 Instance Interaction

The decoder also uses the instance-interaction mechanism associated with DiffusionDet.

Its purpose is to refine candidate instances using their representations and interactions. This is distinct from the spatial alignment operation:

- RiRoI Align determines which visual region a box should inspect.
- Self-attention exchanges information between region features.
- Instance interaction updates the instance representations used for subsequent predictions.

The paper's decoder combines these components rather than relying on box geometry alone.

4.4 Timestep Embedding

The same noisy Gaussian can mean different things at different diffusion timesteps.

For example, a candidate at a high-noise timestep may need substantial correction. A candidate near the final denoising stage may need only a small refinement.

The decoder therefore encodes the timestep t and combines this information with the region and attention features.

Conceptually,

$$\boxed{h_t = \text{TimeEmbedding}(t)}$$

The decoder then uses h_t when predicting the next boxes and their classes.

This lets the network adapt its predictions to the current stage of the reverse process.

4.5 Output Prediction

At the end of a decoder layer, the model predicts boxes and class labels:

$$\boxed{(\hat{B}_t, \hat{C}_t) = D_\theta(z_t, F, t)}$$

The predicted boxes are converted back into Gaussian representations, which can be used in the next denoising step.

During training, predictions are supervised against the ground-truth boxes and labels. During inference, the predicted Gaussian state is updated by the reverse diffusion procedure.

This closes the interaction loop:

Noisy Gaussian candidates
↓
Rotated-region feature alignment
↓
Attention and instance interaction
↓
Box and class prediction
↓
Next denoising state.